FragTree: A Bipartite Graph Framework for Predicting Electron Ionization Mass Spectra of Trimethylsilyl-Derivatized Metabolites

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Abstract

The prediction of electron ionization (EI) mass spectra from molecular structure remains an important unsolved problem in computational metabolomics. Existing tools such as CFM-ID and MassFormer provide useful benchmarks but have limitations in producing full valid spectra. Here we present FragTree, a probabilistic fragmentation modeling framework that represents the fragmentation space of a molecule as a weighted directed acyclic bipartite graph, in which fragment nodes are connected through reaction event nodes. Edge weights representing the conditional probability of each fragmentation event are learned by a graph neural network (GNN) trained on a curated subset of the NIST23 EI mass spectral library. The predicted spectrum for any molecule is constructed by summing path probabilities from the molecular ion to each fragment and normalizing to the most probable fragment, producing a relative intensity vector that can be directly compared against experimental spectra. This manuscript describes the theoretical foundation, graph construction methodology, database curation strategy, and planned evaluation framework for FragTree. Results and comparative evaluation against CFM-ID and MassFormer are pending.

1. Introduction

Electron ionization mass spectrometry (EI-MS) is an analytical technique in metabolomics, environmental chemistry, and forensic science. In EI-MS, molecules are bombarded with a high-energy electron beam (70 eV), resulting in the ejection of an electron and the formation of a radical cation molecular ion (M+). This molecular ion subsequently undergoes a series of fragmentation events producing a characteristic spectrum of fragment ions, the pattern of which encodes structural information about the original molecule. Computational prediction of these spectra from molecular structure alone would have significant practical value: it would enable the annotation of unknown compounds in untargeted metabolomics experiments, facilitate spectral library augmentation for rare or unstable compounds, and provide mechanistic insight into fragmentation chemistry.

Prior computational approaches to EI-MS prediction can be broadly divided into two categories: rule-based fragmentation engines and data-driven machine learning models. Rule-based systems such as the NIST Mass Spectral Fragmentation Simulator encode expert chemical knowledge about common fragmentation pathways, but are limited by the completeness and generalizability of their rule sets. Data-driven approaches have gained prominence in recent years, with CFM-ID (Competitive Fragmentation Modeling for Metabolite Identification) representing one of the most widely used probabilistic frameworks. CFM-ID models fragmentation as a Markov process over a fragmentation graph, learning transition probabilities from training data using an expectation-maximization algorithm. More recently, MassFormer introduced a transformer-based deep learning architecture that learns spectral predictions directly from molecular graph representations without explicitly modeling fragmentation pathways. While both tools demonstrate strong performance on general compound libraries, neither was designed with the specific goal of predicting full spectra and assigning fragment structures to the mass peaks measured.

In this work I present FragTree, a fragmentation prediction framework built around a weighted directed acyclic bipartite graph structure. The key architectural innovation of FragTree relative to prior work is the explicit representation of fragmentation events as nodes in the graph, rather than as edges. This bipartite structure, alternating between fragment nodes and reaction event nodes, allows the model to naturally handle the multi-product nature of bond cleavage events (where a single event produces two fragments), represent distinct reaction types such as bond cleavage, neutral loss, and rearrangement within a unified graph structure, and assign event-specific probability models that can incorporate chemically relevant features. The edge weights in this graph represent conditional probabilities of each event occurring given the parent fragment, and are intended to be learned by a GNN trained on curated EI-MS data from the NIST23 spectral library restricted to confirmed TMS-derivatized compounds.

2. Methods

2.1 Reference Database Construction

The reference spectral library was constructed from the NIST23 EI mass spectral database, accessed via Lib2NIST export. Spectra were exported in .msp format with m/z values binned by adding 0.3 to each value prior to rounding, consistent with standard NIST export practice. A

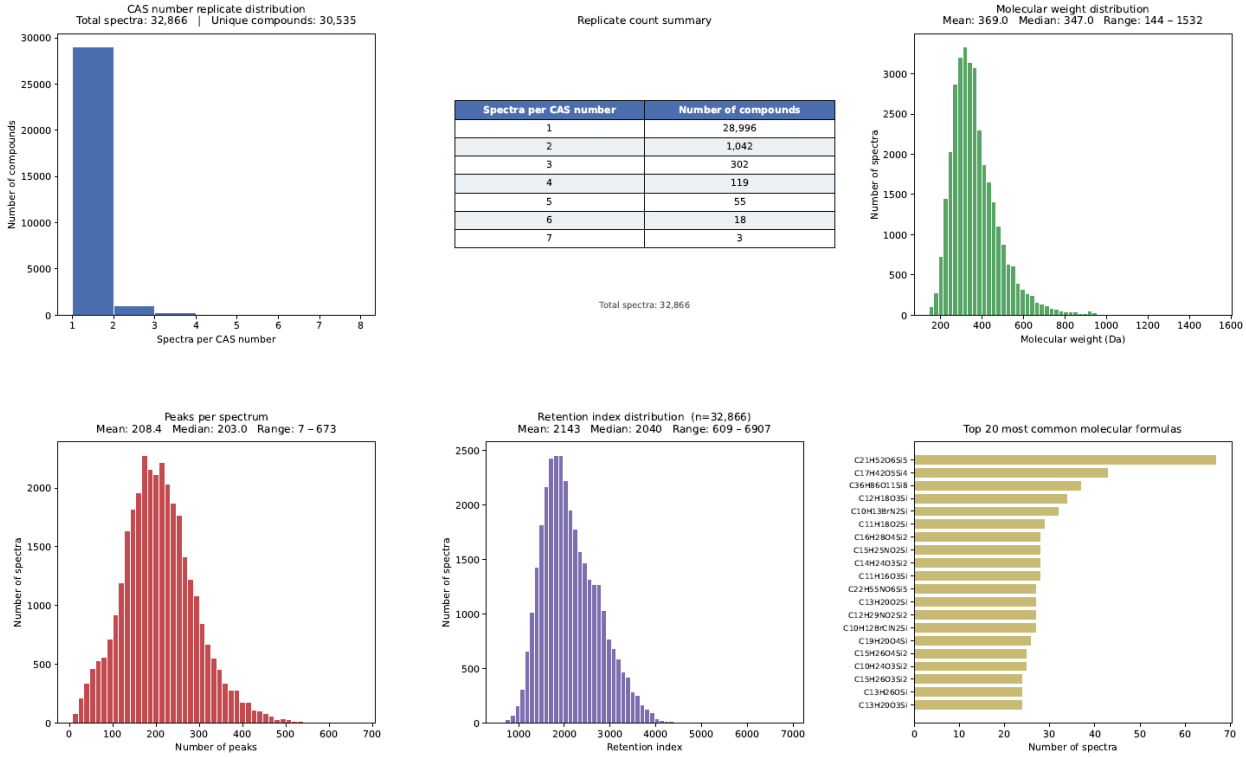
two-stage filtering pipeline was applied to identify confirmed TMS-derivatized spectra. In the first stage, a fragment ion gate required the presence of at least two of the three canonical TMS diagnostic ions (m/z 73, 147, 149) in the spectrum. This gate exploits the chemical necessity of the trimethylsilyl cation fragment: any genuinely TMS-derivatized compound must produce m/z 73 under standard EI conditions. In the second stage, a name-based filter required the compound name to contain either the substring 'tms' or 'trimethylsilyl' (case-insensitive). Requiring both conditions to pass ensures high specificity: the fragment gate eliminates the large majority of non-TMS spectra cheaply before the name check, while the name check removes the small number of false positives that happen to contain m/z 73, 147, or 149 for unrelated chemical reasons. Application of this pipeline to the full NIST23 library yielded 32,866 confirmed TMS spectra representing 30,535 unique compounds as indexed by CAS registry number.

Metadata for each spectrum including compound name, molecular formula, exact mass, molecular weight, InChIKey, and retention index were stored in a SQLite relational database (metadata.db). Spectral data (m/z and intensity arrays) were stored separately in a gzip-compressed HDF5 file (spectra.h5) with each spectrum accessible by a unique key cross-referenced in the metadata table. This separation of metadata and spectral data allows efficient querying of compound properties without loading spectral arrays into memory, and enables scalable storage as the library grows.

Exploratory analysis of the curated library confirmed the expected chemical characteristics of TMS-derivatized metabolites. The molecular weight distribution was approximately normal with a mean of 369 Da and a median of 347 Da, consistent with small molecule metabolites carrying two to six TMS groups. The dominant base peak across the library was m/z 73 (present as the base peak in approximately 33% of spectra), confirming the ubiquity of the trimethylsilyl cation fragment. The mean pairwise cosine similarity between randomly sampled spectra was 0.26, indicating substantial chemical diversity in the dataset and suggesting that clustering-based subsampling will be necessary to obtain a balanced training set.

Molecular Structures were gathered from .mol files also supplied by NIST23. RDKit was used to transform these files to mol objects, which were then converted to binary arrays which were stored in HDF5 files similarly to the spectra. Spectra were mapped to molecular structure data

using a combination of CAS number matching and molecular formula matching, CAS numbers were matched first and then the molecular formula (normalized to Hill notation) was compared between the mol object and the spectra entry in the database.



Spectra per CAS number	Number of compounds
1	28,996
2	1,042
3	302
4	119
5	55
6	18
7	3

Figure 1 - Summary figure for the TMS derived spectra database's metadata. A) CAS number replicate distribution shows the number of replicates in the spectral database. B) Table that shows full values plotted in A. C) Distribution of molecular weights for all included molecular structures. D) Number of peaks detected per spectra. E) distribution of retention index location, used to identify molecules in data collection but not too useful for this project. F) Top 20 most common molecular formulas, different structures sometimes share the same formula, and they may have extremely different structures despite having the same formula.

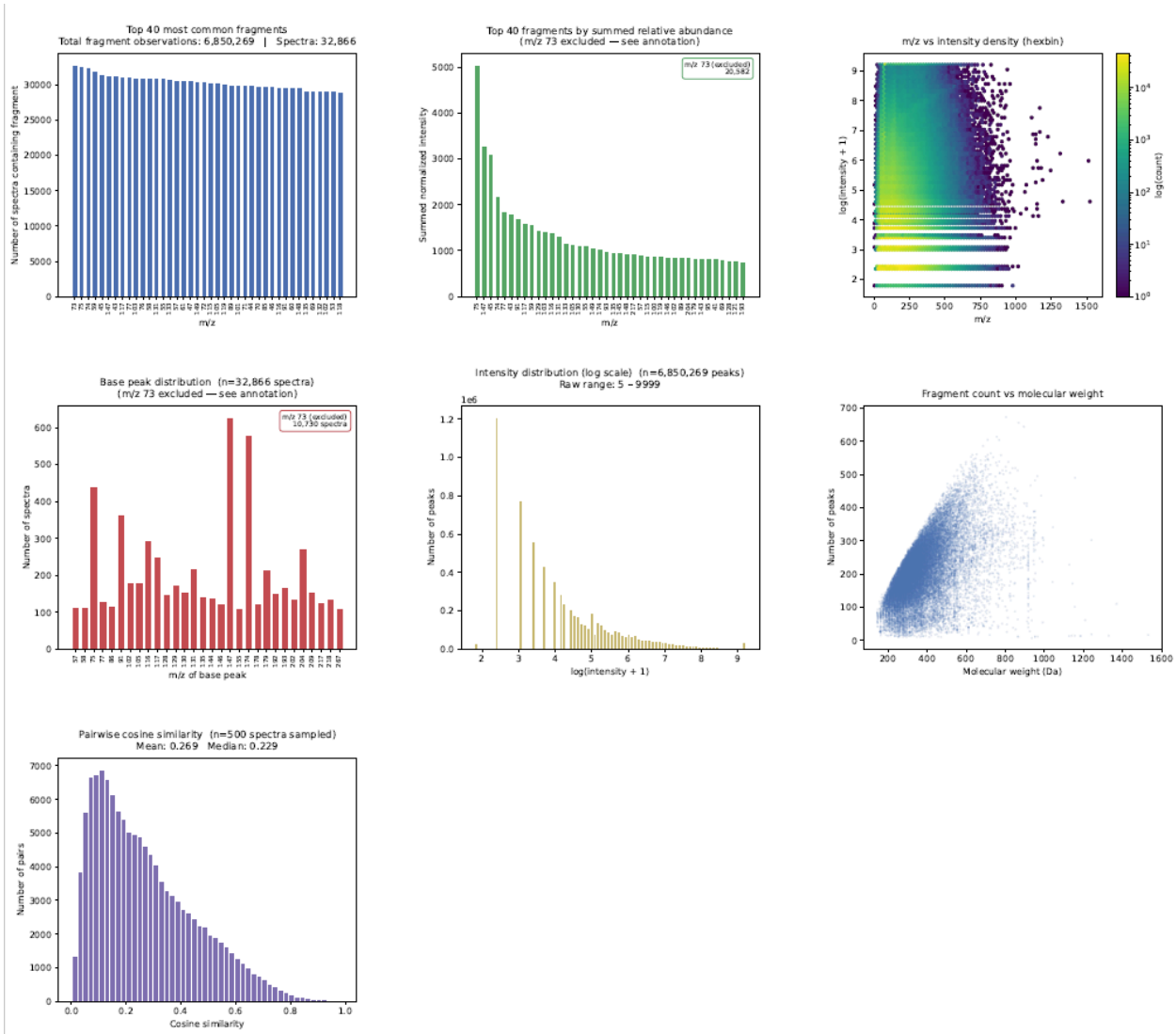


Figure 2 - Summary figure for the spectral data from TMS derived spectra database. A) Top 40 most common fragments shows which fragments are most common, informs tree pruning. B) Top 40 fragments by summed abundance, a fragment may occur very often but at low abundance so this distribution shows that although in A most the fragments are about equally common, their abundances vary greatly. C) M/z vs intensity data shows the relative intensity for each m/z fragment collected. D) Base peak distribution, m/z peaks found to be the most abundant in a given spectra shows very little discernable structure outside of certain peaks being very common. E) Distribution of intensities (log transformed) to show that most intensities are very low. F) Fragment count vs molecular weight shows a scatter plot of how many unique m/z values appear in the spectra of molecules based on molecular weight. G) Cosine similarity between spectra shows the structure of the similarity of the data, giving us a nice somewhat normal distribution and ensuring that we have a reasonably diverse set to move forward with.

2.2 Dataset Clustering and Training Set Selection

Given the size and chemical diversity of the full TMS library, initial model development will be restricted to a structurally coherent subset identified by unsupervised clustering. This approach is motivated by two considerations: first, restricting training to structurally similar compounds reduces the complexity of the fragmentation patterns the model must learn simultaneously; second, it provides a more principled test of the model's ability to generalize within a chemical class before scaling to the broader library.

Molecular structures will be retrieved for each compound using their InChIKey identifiers and represented as Morgan fingerprints (ECFP, radius 2, 2048 bits) computed using RDKit. Prior to clustering, dimensionality reduction will be performed using UMAP (Uniform Manifold Approximation and Projection), which has been shown to preserve both local and global structure in high-dimensional fingerprint spaces more faithfully than linear methods such as PCA. Clustering will then be performed on the UMAP-reduced representations using HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise), a density-based algorithm chosen for its ability to identify clusters of varying density, its explicit handling of outliers as noise points rather than forcing them into clusters, and its freedom from requiring the number of clusters to be specified in advance.

Cluster quality will be assessed using a combination of metrics: the Density-Based Clustering Validation score (DBCV), which is specifically designed for density-based clustering methods and serves as the primary tuning metric; silhouette score, which measures mean intra-cluster cohesion versus inter-cluster separation; cluster persistence from the HDBSCAN condensed tree, which quantifies the stability of each cluster across density thresholds and provides a per-cluster confidence score; noise fraction, which represents the proportion of spectra assigned to the noise class and should remain below approximately 30% to avoid excessive data loss; and Adjusted Rand Index (ARI) and Adjusted Mutual Information (AMI) computed against CAS number groupings as a soft ground truth, since spectra sharing a CAS number should cluster together in a chemically meaningful embedding. The initial training set will be drawn from the densest, highest-persistence cluster identified by this procedure.

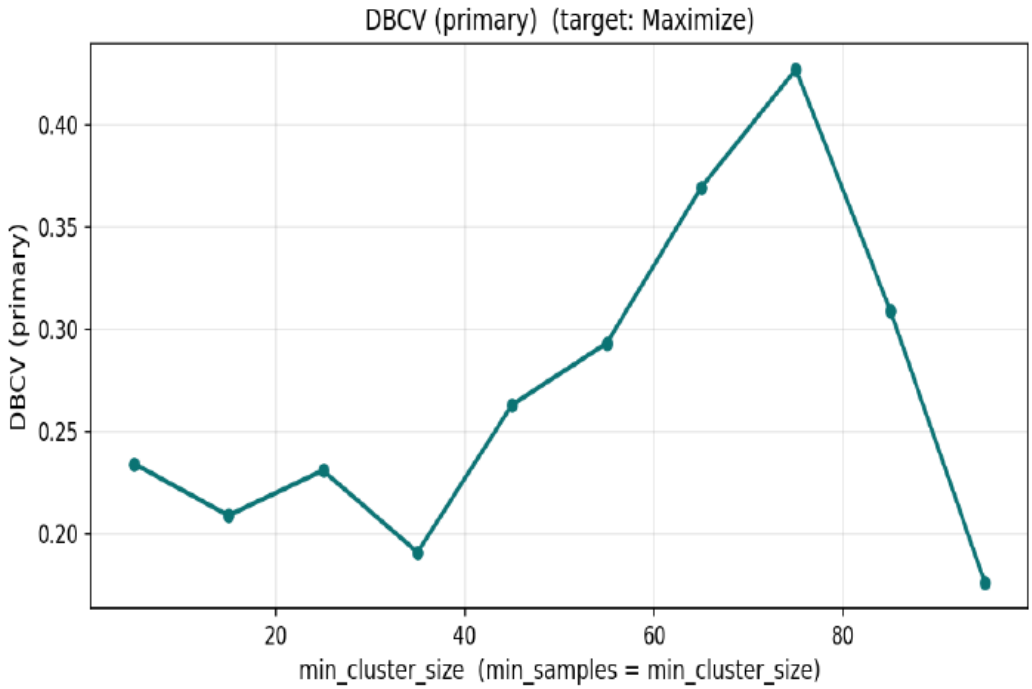


Figure 3 - Simple plot report from coarse grid search of HDBSCAN showing the value of DBCV (primary clustering effectiveness metric) on min_cluster_size = min_samples = [5, 100] step 10.

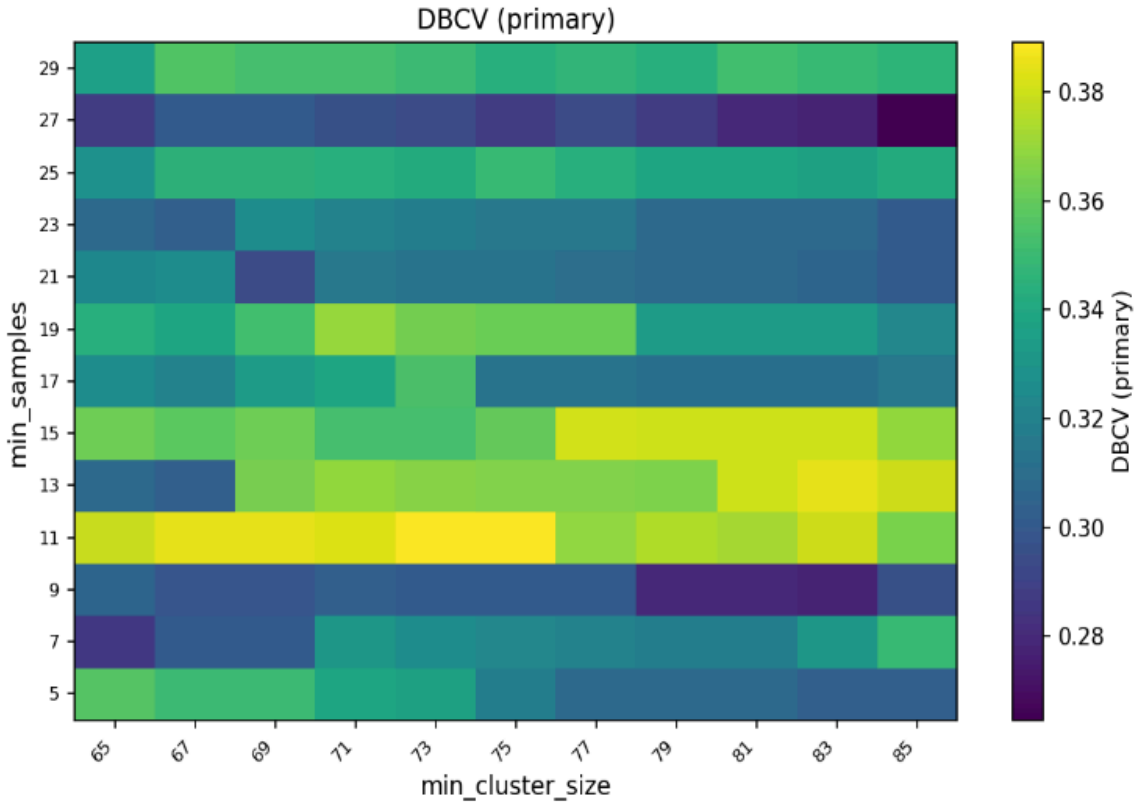


Figure 4 - Heatmap report showing combinations of min_cluster_size and min_samples DBCV scores from a grid search to optimize around ideal parameters found in coarse search

From the results gathered from these searches we determined the ideal values for the parameters min_samples and min_cluster size to be 77 and 11 (CHANGE THIS WE MISREAD ONE OF OUR FIGURES) and ran our final clustering using these values, resulting in a total of 83 clusters with a DBCV of 0.3891 (ALSO CHANGE).

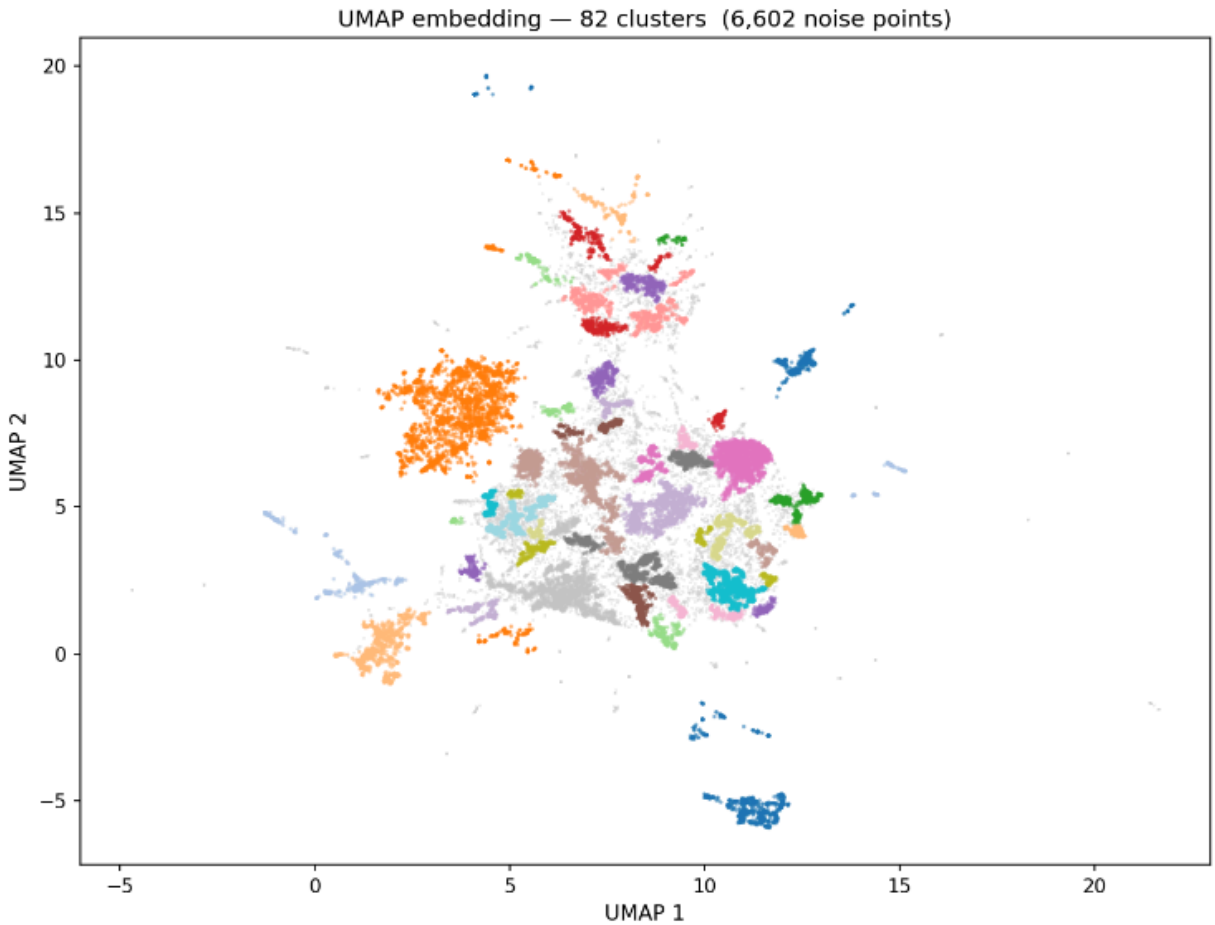


Figure 5 - UMAP embedding of the final clustering output

2.3 Graph Structure

The core data structure of FragTree is a weighted directed acyclic bipartite graph G = (V_F, V_R, E), where V_F is a set of fragment nodes, V_R is a set of reaction event nodes, and E is a set of directed edges connecting fragment nodes to reaction nodes and reaction nodes to fragment nodes. This bipartite construction is motivated by the chemistry of EI fragmentation: a single bond cleavage event typically produces two distinct fragments, and representing the event as a

node rather than as a direct edge between fragments allows both products to be connected to the same event while preserving the directed acyclic structure. Each reaction node in the current implementation represents a bond cleavage event and stores the identity of the cleaved bond (atom indices), the bond order, the location of the retained charge, and the position of the resulting radical.

Fragment nodes represent specific ion structures and store atomic connectivity (as a molecular graph), formal charge, and radical state. The root node of the graph represents the intact molecular ion M+ formed by electron ejection. Fragment generation proceeds recursively: for each fragment node, all cleavable bonds are enumerated, a reaction node is created for each, and the two resulting child fragments are added as new fragment nodes connected to that reaction node. Recursion terminates when a fragment falls below a minimum mass threshold of 100 Da, ensuring that highly abundant but structurally uninformative small fragments (which are shared across essentially all TMS-derivatized molecules) do not dominate the graph or the predicted spectrum. Hydrogen atoms are excluded from the fragmentation enumeration, as the combinatorial explosion of hydrogen rearrangement pathways would make the graph computationally intractable for molecules of practical size.

Each directed edge e from a fragment node f_parent to a reaction node r carries a weight w(e) representing the conditional probability that reaction r occurs given that fragment f_parent has formed. Each directed edge from a reaction node r to a child fragment node f_child carries a complementary weight representing the probability that f_child is the charged (measured) product of that reaction as opposed to the neutral unmeasured product. The full probability of a fragment f forming is computed as the sum over all paths from the root to f of the product of edge weights along each path. Predicted relative intensities are obtained by normalizing all fragment probabilities by the probability of the most probable fragment, yielding a normalized intensity vector that is directly analogous to an experimental mass spectrum.

2.4 Edge Weight Prediction

Edge weights will be predicted by a graph neural network in which each bond in the parent fragment receives its own dedicated embedding. This bond embedding is constructed by concatenating two components: a global molecular fingerprint and a set of local features specific to the bond and its two constituent atoms.

The global component of the bond embedding is a Morgan fingerprint (ECFP, radius 2, 2048 bits) computed for the parent fragment as a whole using RDKit, identical to the fingerprint used for dataset clustering. This encodes the broader structural context of the molecule in which the bond sits. The local component concatenates features of the bond itself with features of each of the two atoms it connects, producing a fixed-length vector that captures the immediate chemical environment of the candidate cleavage site.

The bond-level features included are: bond length (estimated from covalent radii or obtained from a 3D conformer), bond order (single, double, or triple), whether the bond is part of a ring system, whether the bond is conjugated, and bond polarity estimated as the absolute difference in Pauling electronegativity between the two bonded atoms. The per-atom features included for each of the two atoms forming the bond are: atomic number, formal charge, degree (number of bonded neighbors), valence shell occupancy (number of electrons in the outermost shell), hybridization state (sp, sp2, sp3, or other), whether the atom is aromatic, electronegativity (Pauling scale), covalent radius, whether the atom is part of a ring system, and the number of attached hydrogen atoms.

Following bond cleavage, the resulting radical must be assigned a location in the fragment structure. A localized treatment of the radical is adopted here, as it has been shown to yield more accurate fragmentation predictions than delocalized models by allowing downstream bond cleavage probabilities to be explicitly conditioned on radical proximity. When a bond between atoms i and j is cleaved to produce the charged fragment, the radical is assigned with equal prior probability (0.5) to either atom i or atom j. This 50/50 prior reflects the absence of strong a priori grounds for preferring one terminus over the other before resonance or inductive stabilization effects are considered. The radical position is then encoded as a binary flag in the atom feature vector, propagated through all subsequent bond evaluations within that fragment so that the model can learn how radical proximity influences lability at each site.

The complete bond embedding (global Morgan fingerprint concatenated with local bond and atom features) is passed through a two-layer multilayer perceptron with a sigmoid output activation to produce the conditional cleavage probability for that bond given the parent

fragment. This is applied independently to every bond in the fragment, scoring all candidate cleavage events in a single forward pass.

The model will be trained to minimize the cosine distance between the predicted normalized intensity vector and the experimentally observed spectrum for each training compound. Because many fragments in the graph correspond to intermediate species that carry no charge and are therefore not directly observed by the mass spectrometer, the training signal propagates through the graph from measured endpoint fragments back through unmeasured intermediates. This is analogous to the treatment of latent variables in the CFM-ID framework, though FragTree uses gradient-based optimization rather than the expectation-maximization approach employed by CFM-ID.

2.5 Evaluation Framework

FragTree will be evaluated against two state-of-the-art competing methods: CFM-ID 4.0 and MassFormer. CFM-ID 4.0 is a probabilistic fragmentation modeling tool that uses competitive fragmentation modeling with learned transition probabilities; it represents the most widely used dedicated EI-MS prediction tool in the metabolomics community. MassFormer is a more recent transformer-based approach that predicts spectra directly from molecular graph representations without explicit fragmentation modeling, representing the current state of the art in deep learning-based spectral prediction.

Performance will be quantified using a battery of metrics applied to held-out test spectra from the TMS library. The primary metric will be cosine similarity between predicted and experimental spectra, as this is the most widely reported metric in the spectral prediction literature and is directly used in library search tools. Cosine similarity is computed as the dot product of the predicted and experimental intensity vectors (each normalized to unit length), where the vectors are aligned by m/z value. Supplementary metrics will include the weighted cosine similarity (which weights each m/z contribution by the square root of its intensity to reduce the influence of dominant fragments), top-k peak recall (the fraction of the k most intense experimental peaks that are also among the k most intense predicted peaks, evaluated at k = 5, 10, and 20), mean absolute error of predicted intensities for matched peaks, and the proportion of test spectra achieving cosine similarity above threshold values of 0.7, 0.8, and 0.9, as these thresholds are commonly used to define a confident spectral match in library search applications.

All methods will be evaluated on identical train-test splits stratified by CAS number to prevent data leakage.

3. Results

[To be completed following model training and evaluation.]

4. Discussion

[To be completed following model training and evaluation.]

References

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